Zhengzhong Ricky You

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Research Interests

- Exact algorithms for large-scale integer and combinatorial optimization, with emphasis on branch-price-and-cut, decomposition, and extended formulations.
- Learning-augmented optimization, decision-focused learning, reinforcement learning for combinatorial optimization, and interfaces between operations research and machine learning.
- Large language models for healthcare operations, including routing, scheduling, and resource allocation under uncertainty and fairness constraints.

Education

- **University of Florida**, M.S. in Operations Research, Industrial & Systems Engineering; GPA: 3.95/4.00 **2021–2025**
- **Nankai University**, B.S. in Chemistry; GPA: 3.67/4.00 **2017–2021**

Publications

[J1] *Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly*.
Zhengzhong Ricky You, Yu Yang, Xinshang Wang, Wotao Yin. **Operations Research**.

[J2] *RouteOpt: An Open-Source Modular Exact Solver for Vehicle Routing Problems*.
Zhengzhong Ricky You, Yu Yang. **INFORMS Journal on Computing**.

[C1] *Draft-and-Audit Reinforcement Learning for Optimization Modeling*.
Zeping Min, Weihang Xu, **Zhengzhong Ricky You**, Wotao Yin, Xinshang Wang. **Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)**, PMLR 306, 2026.

[S1] *Learned Pairwise Deep Dual-Optimal Inequalities for Stabilizing Column Generation*.
Zhengzhong Ricky You, Bo Tang, Haoran Liu, Baichuan Mo. Under review at **INFORMS Journal on Computing**.

[S2] *Bulk Service Queueing for Transit Resilience under Short Random Service Suspensions*.
Baichuan Mo, Li Jin, **Zhengzhong Ricky You**, Zuo-Jun Max Shen, Haris N. Koutsopoulos, Jinhua Zhao. Under review at **Transportation Science**.

Research Experience

- **TikTok Inc.** **2026**
  Machine Learning Engineer Intern, E-commerce Supply Chain & Logistics *San Jose, CA*
  - Conducted research on inventory control for e-commerce supply chains under the mentorship of Baichuan Mo, Yikang Hua, and Zhou Ye.
- **MIT-UF-NEU Joint Summer Research Camp** **2026**
  Selected Participant, MIT JTL-Transit Lab program
  - Studied large-scale inventory placement under demand uncertainty with mentorship from Baichuan Mo.
- **University of Kentucky** **2026**
  Visiting Scholar, Department of Statistics
  - Conducting research on online learning and branch-and-bound algorithms.
- **DAMO Academy, Alibaba Group** **2025–2026**
  Research Intern *Bellevue, WA*
  - Studied large-language-model-assisted solver parameter tuning under the mentorship of Xinshang Wang and Wotao Yin. The tuned configurations improved average benchmark performance over solver defaults; the tested black-box baselines generally did not.

Research Software

- **RouteOpt** — **2021–Present**
  Lead Developer, 20k+ LOC in C++ — *Open source*
  - Designed and implemented an open-source exact solver for the CVRP and CVRPTW, with modular branching, cutting-plane, and variable-reduction components, learning-augmented branching, and reproducible experiment scripts.
  - On standard benchmarks, RouteOpt nearly doubled the speed of VRPSolver and proved optimality for four previously open instances; one required approximately 1.5 years of single-core time.
- **TorchDCM** — **2026**
  Co-developer, Python — *Open source*
  - Co-developed a unified PyTorch-native package for discrete choice modeling, with vectorized likelihoods, automatic differentiation, multiple logit model families, and standard econometric outputs.

Conference & Workshop Presentations

- *L-DDOIs: Learning-Based Deep Dual-Optimal Inequalities for Stabilizing Column Generation*, INFORMS Transportation Science and Logistics Conference, MIT Sloan — **2026**
- *Two-Stage Learning to Branch in Branch-Price-and-Cut Algorithms for Solving Vehicle Routing Problems Exactly*, invited talk, UK AI and ML Symposium and Nontechnical Workshop — **2025**
- *Accelerating Dynamic Programming via Dual Selection for Column-Generation Frameworks*, INFORMS Optimization Society Conference — **2024**
- INFORMS Annual Meeting: L-DDOIs (2025); RouteOpt (2024); Two-Stage Learning to Branch (2023); Learning to Branch with Column Generation (2022) — **2022–2025**

Awards & Honors

- Harold D. Haldeman, Jr. Graduate Fellowship, University of Florida — **2024**
- INFORMS Optimization Society Conference Travel Support — **2024**
- Academic Excellence Scholarship; Innovation and Entrepreneurship Scholarship, Nankai University — **2019–2020**
- First Prize, Tianjin Theoretical Chemistry Competition; First Prize, Tianjin Mathematics Competition — **2018–2019**

Teaching & Mentoring

- **Sole Instructor**, ESI 3327C Matrix & Numerical Methods in Systems Engineering, University of Florida; 52 undergraduates; linear algebra, numerical methods, and introductory optimization — **2024**
- **Outreach Mentor**, UF Student Science Training Program; mentored Ken Cheng and Shrey Gupta

Professional Service

Journal referee for *INFORMS Journal on Computing*, *INFORMS Journal on Optimization*, *Transportation Science*, *Transportation Research Part B*, *IISE Transactions*, *Computers & Operations Research*, and *OR Spectrum*.

Selected Relevant Courses

ISE Ph.D. level (GPA: 4.00): Discrete Optimization; Linear Programming and Network Optimization; Stochastic Modeling & Analysis; Advanced Topics in Continuous Optimization. **Mathematics Ph.D. level (GPA: 4.00)**: Analysis I; Numerical Linear Algebra; Numerical Analysis; Numerical Optimization. **CS Ph.D. level (GPA: 4.00)**: Machine Learning.

Skills

Methods: branch-price-and-cut, column generation, dual stabilization, cutting planes, learning-augmented optimization, stochastic and robust optimization. **Programming**: C++, Python (PyTorch and reinforcement-learning libraries), MATLAB, Shell, Git, Linux. **Software**: RouteOpt, TorchDCM, Gurobi, CPLEX, SCIP, reproducible experiment pipelines.